A balanced 13-variable Boolean function with absolute indicator 112

Claude (AI model by Anthropic)
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Abstract

We exhibit an explicit balanced Boolean function $f\colon \mathbb{F}_2^{13} \to \mathbb{F}_2$ of algebraic degree 11 whose absolute indicator is $\Delta_f = \max_{d\neq 0}\left|\sum_x (-1)^{f(x)\oplus f(x\oplus d)}\right| = 112$. Since $112 < 2^{\lfloor (13+1)/2\rfloor} = 128$, this is a counterexample to the conjecture of Zhang and Zheng (1995) in the case $n = 13$; to our knowledge, for odd n the conjecture had previously been disproved only for $n \in \{9, 11, 15, 21\}$. The function is invariant under the affine group $\mathrm{AGL}(1, 13)$ permuting the coordinates and was found by a local search in the 2^{74}-element class of such functions. Balancedness, the degree bound $\deg f \geq 3$ and $\Delta_f \leq 112$ are certified in Lean 4 (core library only), with all computations checked by the kernel and only the standard axioms.

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1 Introduction

For a Boolean function $f\colon \mathbb{F}_2^n \to \mathbb{F}_2$ the *autocorrelation* at $d \in \mathbb{F}_2^n$ is $r_f(d) = \sum_{x\in\mathbb{F}_2^n} (-1)^{f(x)\oplus f(x\oplus d)}$, and the *absolute indicator* is $\Delta_f = \max_{d\neq 0} |r_f(d)|$. Together with the sum-of-squares indicator, Δ_f was introduced by Zhang and Zheng [19] as a measure of global avalanche characteristics (GAC);

a small Δ_f means that every nontrivial derivative $x \mapsto f(x) \oplus f(x \oplus d)$ is close to balanced. Bent functions have $\Delta_f = 0$ but are never balanced, and the problem of how small Δ_f can be for a balanced function has been studied for three decades. The original statement [19, Conjecture 1, p. 329] reads: *"Let f be a balanced function on V_n, whose algebraic degree is at least three. Then $\Delta_f \geq 2^{\frac{1}{2}(n+1)}$."* In the recent literature (we follow Tang and Maitra [17, Conjecture 1]) the exponent is written with a floor, which gives a meaningful statement for even n as well; for odd n, in particular for $n = 13$, the two forms coincide:

Conjecture 1.1 (Zhang–Zheng)**.** *The absolute indicator of any balanced Boolean function f on n variables of algebraic degree at least 3 satisfies $\Delta_f \geq 2^{\lfloor (n+1)/2 \rfloor}$.*

Prior counterexamples. For odd n the conjecture was disproved for $n = 15$ by modifications of Patterson–Wiedemann functions (Maitra–Sarkar [14], a balanced function with $\Delta_f = 216 < 256$), for $n = 21$ by Patterson–Wiedemann-type functions (Gangopadhyay–Keskar–Maitra [5]; the absolute indicator computed there was corrected, and twenty further 21-variable examples were given, by Kavut [6], see also [16]), and for $n = 9$ ($\Delta_f = 24$) and $n = 11$ ($\Delta_f = 56$) by Kavut, Maitra and Yücel through a steepest-descent-like search among rotation-symmetric functions [8, 9]. For even n the conjecture reads $\Delta_f \geq 2^{n/2}$ and was disproved for $n = 14$ by Burnett, Millan, Dawson and Clark [1], for $n = 10$ in [9], for $n \equiv 2 \pmod 4$, $n \geq 46$ by the construction of Tang and Maitra [17], and subsequently for all even $n \geq 12$ by constructions and searches [7, 18, 15]. The list "$n = 9, 11, 15, 21$" for odd n is given as the state of the art in the introduction and in the concluding open problems of [17], in [9, Sec. 1], and in the survey talk [16] (slide "Some counterexamples on Zhang–Zheng conjecture", listing $n = 9, 10, 11, 14, 15, 21$), and the same odd list $n \in \{9, 11, 15, 21\}$ is repeated in the 2019 survey of known results by Canteaut, Kölsch and Wiemer [2, Sec. 5]; the later papers [7, 18, 15] treat even n only. The case $n = 13$ is the smallest odd case left open. For $n = 13$ the conjectured value 128 is attained, e.g., by the construction of Zhang and Zheng themselves on V_{2k+1} [19, Theorem 17], while the balanced 13-variable functions of record nonlinearity 4036 of Kavut and Yücel have $\Delta_f = 536$ [11, Appendix B]; we are not aware of any balanced 13-variable function with $\Delta_f < 128$ in the literature.

Result. We settle the case $n = 13$.

Theorem 1.2. *Let f be the Boolean function on 13 variables whose truth table is given in Appendix A (equivalently, the union of the 36 $\mathrm{AGL}(1, 13)$-orbits listed in Appendix B). Then f is balanced, $\deg f = 11$, and $|r_f(d)| \leq 112$ for every $d \neq 0$, with equality for 312 values of d. In particular $\Delta_f = 112 < 128 = 2^{\lfloor (13+1)/2 \rfloor}$, so Conjecture 1.1 fails for $n = 13$.*

The parts "balanced", "$\deg f \geq 3$" and "$\Delta_f \leq 112$" are certified in Lean (Section 4); the exact degree 11, the equality count and the nonlinearity 3980 quoted below are checked by an independent script only. To our knowledge, as of October 8, 2026, no balanced 13-variable Boolean function with $\Delta_f < 128$ had been reported.

2 Definitions and elementary facts

We identify $x = (x_0, \ldots, x_{12}) \in \mathbb{F}_2^{13}$ with the integer $\sum_i x_i 2^i \in \{0, \ldots, 2^{13} - 1\}$; then $x \oplus d$ is the bitwise exclusive or. A function f is given by the integer $\mathrm{tt} = \sum_x f(x) 2^x < 2^{8192}$ (its truth table). It is *balanced* if $\mathrm{wt}(f) = |\{x : f(x) = 1\}| = 2^{12}$. The algebraic normal form is $f(x) = \bigoplus_u a_u \prod_{i : u_i = 1} x_i$ with Möbius coefficients $a_u = \bigoplus_{x \preceq u} f(x)$, where $x \preceq u$ means $x \wedge u = x$

bitwise, and $\deg f = \max\{\mathrm{wt}(u) : a_u = 1\}$. The Walsh transform is $W_f(a) = \sum_x (-1)^{f(x)\oplus a\cdot x}$, the nonlinearity is $2^{n-1} - \frac{1}{2}\max_a |W_f(a)|$.

Lemma 2.1 (Wiener–Khinchin)**.** *$r_f(d) = 2^{-n}\sum_a W_f(a)^2(-1)^{a\cdot d}$ for all d, and $\sum_a (W_f(a)^2 - 2^n)^2 = 2^n \sum_{d\neq 0} r_f(d)^2$.*

Proof. $\sum_a W_f(a)^2(-1)^{a\cdot d} = \sum_{x,y}(-1)^{f(x)\oplus f(y)}\sum_a (-1)^{a\cdot(x\oplus y\oplus d)} = 2^n \sum_x (-1)^{f(x)\oplus f(x\oplus d)}$. By Parseval for the pair (r_f, W_f^2) this gives $\sum_a W_f(a)^4 = 2^n \sum_d r_f(d)^2$, and with $\sum_a W_f(a)^2 = 2^{2n}$, $r_f(0) = 2^n$ the second identity follows by expanding the square. □

Lemma 2.2. *If $n \geq 3$ and $\mathrm{wt}(f)$ is even (in particular if f is balanced), then $r_f(d) \equiv 0 \pmod 8$ for all d.*

Proof. For $d \neq 0$ let $S = \{x : f(x) = 1\}$. Then $r_f(d) = 2^n - 2\,\mathrm{wt}(D_d f)$ with $\mathrm{wt}(D_d f) = |S \triangle (S \oplus d)| = 2|S| - 2|S \cap (S \oplus d)|$. The set $S \cap (S \oplus d)$ is invariant under the fixed-point-free involution $x \mapsto x \oplus d$, so its size is even; hence $\mathrm{wt}(D_d f) \equiv 2|S| \equiv 0 \pmod 4$ and $r_f(d) \equiv 2^n \equiv 0 \pmod 8$. □

Thus for balanced f on 13 variables the first admissible value below the conjectured bound 128 is 120. We remark also that concatenating two 12-variable bent functions $g_0 \| g_1$ always gives $\Delta \geq 128$: here $r(d,1) = 2\sum_x (-1)^{g_0(x)\oplus g_1(x\oplus d)} = 2W_{\tilde{g}_0\oplus\tilde{g}_1}(d)$ (with $\tilde{g}_i$ the dual bent functions), and $\max_d |W_h(d)| \geq 2^6$ for every 12-variable h.

3 Construction and search

Symmetry class. Let $G = \mathrm{AGL}(1,13) = \{i \mapsto ai + b \bmod 13 : a \in \mathbb{Z}_{13}^\times,\ b \in \mathbb{Z}_{13}\}$ (order 156) act on $\mathbb{F}_2^{13}$ by permuting coordinates, $(\pi x)_{\pi(i)} = x_i$. It is generated by the cyclic shift $i \mapsto i+1$ and the map $i \mapsto 2i$ (as 2 is a primitive root mod 13); it contains the dihedral group, so G-invariant functions are in particular rotation- and dihedral-symmetric. The group has 74 orbits on $\mathbb{F}_2^{13}$ (sizes $1, 13, 26, 39, 52, 78, 156$), so there are 2^{74} invariant functions. This class appears already in [12], where Kavut and Yücel used it (as dihedral-symmetric functions with the extra permutation $x_i \mapsto x_{2i}$) to search for highly nonlinear 13-variable functions.

Orbit formulas. Let $O_1, \ldots, O_{74}$ be the orbits with representatives $u_j \in O_j$ and put $C_{ij} = \sum_{x\in O_i}(-1)^{u_j\cdot x}$. Since permutation matrices are orthogonal, $W_f(\pi a) = W_f(a)$ and $r_f(\pi d) = r_f(d)$ for G-invariant f. Writing $s_i = (-1)^{f(O_i)}$ we get

$$W_f(u_j) = \sum_i C_{ij} s_i, \qquad r_f(u_j) = 2^{-13}\sum_i W_f(u_i)^2\, C_{ij}$$

(the second formula is Lemma 2.1 grouped by orbits). Hence the full Walsh spectrum and autocorrelation of an invariant function are computed from a 74×74 integer matrix.

Search. We used the cost function of Kavut and Yücel [10, 9], $\mathrm{cost}(f) = \sum_a (W_f(a)^2 - 2^{13})^2 = 2^{13}\sum_{d\neq 0} r_f(d)^2$ (Lemma 2.1), with moves that complement f on one orbit, steepest descent with a tabu list of length 5, 3000 steps per run and random restarts; after each step Δ_f and the presence of a zero Walsh coefficient were recorded. (A zero Walsh coefficient $W_f(a) = 0$ means that $f \oplus a\cdot x$ is balanced with the same Δ; the function below is itself balanced.) Each step costs $O(74^2)$ operations, so thousands of restarts take seconds. In a run with seed 41 the first function

with a Walsh zero and $\Delta \le 120$ appeared after 259 restarts (it is balanced, with $\Delta = 120$ and nonlinearity 3984); the function of Theorem 1.2 (with $\Delta = 112$) was found after 1015 restarts with the seed reported below, in about 40 seconds. Short exploratory runs in the larger classes generated by the shift and $i \mapsto ai$ for a of order $2, 3, 4, 6$ (380, 232, 198, 140 orbits) did not get below $\Delta = 136$; a search for $\Delta \le 104$ in $\mathrm{AGL}(1, 13)$ found further functions with $\Delta = 112$ but none with $\Delta \le 104$.

Reproduction. With the files in `data/search/`:

```
python3 t1.py && python3 export.py agl13          # orbits of AGL(1,13), 74x74 matrix
gcc -O3 -march=native -o acls4 acls4.c -lm
./acls4 agl13.bin 13 112 3000 77 100000 5   # class n target steps seed restarts tabu
```

It prints a line beginning `r1015 it2098 ZERO Delta=112 maxW=232 s=`, followed by a 74-bit string giving the value of f on the orbits ordered by their least element; this string is stored in `data/search/orbit_string_agl13.txt` and is equivalent to Appendix B.

The certificate. The truth table is the hexadecimal integer in Appendix A. The file `data/truth_table.hex` contains its 2048 lower-case hex digits followed by a newline; its SHA-256 hash is

`0b4a77a6d9278877e3f206da6bf4c185bbe29ed76f50726e30843e5a97dba0c8`

(the same digits are the literal `tt` in `lean/Main.lean` and Appendix A). Bit x of the integer is $f(x)$. The script `data/verify.py` recomputes, by the naive $O(4^{13})$ double sum, the weight 4096, all 8191 values $r_f(d)$, the Walsh spectrum, the algebraic normal form and the G-invariance. Its output: $\deg f = 11$ (and $a_{\{0,1,2\}} = 1$), $\max_a |W_f(a)| = 232$, i.e. nonlinearity 3980 (script-checked only), and the distribution of $|r_f(d)|$, $d \neq 0$:

$\lvert r_f(d)\rvert$	0	8	16	24	32	40	48	56	64	72	80	88	96	104	112
count	156	416	780	1392	650	923	416	780	156	585	286	936	247	156	312

All counts except the one for 24 are multiples of 13, as they must be for a function invariant under the cyclic shift: the only shift-fixed nonzero vector is the all-ones vector $\mathbf{1}$, and $r_f(\mathbf{1}) = 24$. The ANF of f has 4485 monomials.

Proof of Theorem 1.2. The statement concerns one explicit function, so the proof is the finite computation; it was carried out independently by the script above and, for balancedness, $\deg f \ge 3$ and $\Delta_f \le 112$, by the Lean kernel as described in Section 4. □

4 Formal verification

The file `lean/Main.lean` (374 lines) is checked by Lean 4.34.1 [4] using only the core library (no Mathlib). There is no `sorry`, no `axiom` and no `native_decide`; all evaluations use `decide +kernel`.

4.1 Statement

The definitions used by the statement are, verbatim (`tt` is the 2048-digit hexadecimal literal of Appendix A, abbreviated here):

```
def tt : Nat := 0xffeafdcdeba3a0e7...4f1713477dfe4eb75bfea7de3e68
def f (x : Nat) : Bool := tt.testBit x
def sgn (b : Bool) : Int := if b then -1 else 1
def autocorr (d : Nat) : Int := ((List.range (2^13)).map (fun x => sgn (f x ^^ f (x ^^^ d)))).sum
def weight : Nat := ((List.range (2^13)).filter f).length
def anfCoeff (u : Nat) : Bool :=
  ((List.range (2^13)).filter (fun x => x &&& u == x)).foldl (fun b x => b ^^ f x) false
def monoDeg (u : Nat) : Nat := ((List.range 13).filter (fun i => u.testBit i)).length

def MainStatement : Prop :=
  weight = 2^12 ∧
  (∃ u, u < 2^13 ∧ 3 ≤ monoDeg u ∧ anfCoeff u = true) ∧
  (∀ d, 0 < d → d < 2^13 → (autocorr d).natAbs ≤ 112)
```

The certified theorems are

```
theorem main :
    weight = 2^12 ∧
    (∃ u, u < 2^13 ∧ 3 ≤ monoDeg u ∧ anfCoeff u = true) ∧
    (∀ d, 0 < d → d < 2^13 → (autocorr d).natAbs ≤ 112)
theorem main_statement : MainStatement := main
theorem zhang_zheng_fails_n13 :
    ∃ g : Nat → Bool,
      ((List.range (2^13)).filter g).length = 2^12 ∧
      (∃ u, u < 2^13 ∧ 3 ≤ monoDeg u ∧
        ((List.range (2^13)).filter (fun x => x &&& u == x)).foldl (fun b x => b ^^ g x) false =
    true) ∧
      (∀ d, 0 < d → d < 2^13 →
        (((List.range (2^13)).map (fun x => sgn (g x ^^ g (x ^^^ d)))).sum).natAbs < 2^((13+1)/2))
```

`MainStatement` is placed in the file immediately after the definitions; the identity `main_statement` guarantees that it is literally the statement proved at the end.

4.2 Statement audit

- *Encoding.* Inputs are naturals $x < 2^{13}$ whose bit i is x_i; `x ^^^ d` is bitwise xor, i.e. translation by d in $\mathbb{F}_2^{13}$; `List.range (2^13)` enumerates $\mathbb{F}_2^{13}$ exactly once. Only bits $x < 2^{13}$ of `tt` are ever read (and $\mathrm{tt} < 2^{8192}$ anyway).

- *Autocorrelation.* `autocorr d` is literally $\sum_x (-1)^{f(x)\oplus f(x\oplus d)}$ (`sgn true = -1`); the quantifier ranges over all d with $0 < d < 2^{13}$, i.e. all nonzero shifts, and `natAbs` is the absolute value. In `zhang_zheng_fails_n13` the inequality is strict and `(13+1)/2` is natural-number division, equal to $7 = \lfloor 14/2 \rfloor$.

- *Balanced.* `weight = 2^12` says exactly half of the 8192 inputs are mapped to 1.

- *Degree.* `anfCoeff u` is the Möbius coefficient $a_u = \bigoplus_{x \preceq u} f(x)$ (with $x \preceq u \iff x \wedge u = x$), the coefficient of $\prod_{i\in u} x_i$ in the ANF, and `monoDeg u` is the number of variables of that monomial; so the second conjunct is exactly $\deg f \geq 3$ (the witness is $u = 7$, the monomial $x_0x_1x_2$).

- The auxiliary functions below never occur in the statements.

4.3 From the naive definitions to a kernel-feasible computation

A literal evaluation of `autocorr` for all d needs 2^{26} list steps, far beyond what the kernel can do (a test with 2^{18} steps already exhausted 8 GB). The proof therefore evaluates a fast version built from `Nat` bitwise operations, which the Lean kernel executes with GMP, and proves it equal to the naive definitions by general lemmas:

1. `sum_sgn`: for any list L and predicate p, `(L.map (fun x => sgn (p x))).sum = L.length - 2 * (L.filter p).length`. Hence `autocorr d` $= 8192 - 2N_d$ with $N_d = |\{x : f(x) \neq f(x \oplus d)\}|$.

2. *Translation by block swaps.* With masks `M i` (ones exactly at positions $x < 2^{13}$ whose bit i is 0) and `swapBlk i U := ((U >>> 2^i) &&& M i) ||| ((U &&& M i) <<< 2^i)`, lemma `swapBlk_testBit` proves `(swapBlk i U).testBit x = U.testBit (x ^^^ 2^i)` for all U, $i < 13$, $x < 2^{13}$. Its proof is a case analysis on bit i of x using the core lemmas on `testBit` of `&&&`, `|||`, `>>>`, `<<<`, together with $13 \cdot 8192$ elementary facts about the masks and about $x \oplus 2^i$ (`factOK`), checked by `decide +kernel` in 13 separate theorems. Composing the swaps for the bits of d gives `perm d T` with `perm_testBit : (perm d T).testBit x = T.testBit (x ^^^ d)` for $d, x < 2^{13}$ (induction over the bits, plus a kernel check that the accumulated xor mask equals d for all $d < 2^{13}$).

3. *Population count modulo* $2^{14} - 1$. Let $D_q = \sum_{j<q} 2^{14j}$ and $Y(V, q) = \sum_{t=0}^{13}((V \gg t) \wedge D_q)$. Lemma `and_D_succ` ($A \wedge (2^{14}E + 1) = 2^{14}((A \gg 14) \wedge E) + (A \bmod 2)$) gives `Y_succ`: $Y(V, q+1) = 2^{14}Y(V \gg 14, q) + \#\{t < 14 : V_t = 1\}$, and since $2^{14} \equiv 1 \pmod{2^{14}-1}$, induction gives `Y_mod`: $Y(V, q) \equiv \#\{x < 14q : V_x = 1\} \pmod{16383}$. With $q = 586$ ($14q = 8204$) and the low 8192 bits masked, `popc_eq : popc V = cntBits V 8192` where `popc V := Y (V &&& (2^8192-1)) 586 % 16383` and `cntBits V m` is the naive count of set bits below m.

4. `autocorr_eq`: for $d < 2^{13}$, `autocorr d = 8192 - 2 * popc (tt ^^^ perm d tt)`, from 1–3.

5. `chk0`,…,`chk7`: the kernel verifies $4040 \leq$ `popc (tt ^^^ perm d tt)` ≤ 4152 for all $0 < d < 2^{13}$, in eight chunks of 1024 shifts; each shift costs roughly a hundred operations on 8192-bit integers. Together with 4 this is $|r_f(d)| \leq 112$.

6. `weight_eq`: `weight = popc tt = 4096`; and `anfCoeff 7 = true`, `monoDeg 7 = 3`, both by `decide +kernel`.

4.4 Axioms and resources

The output of `lean Main.lean`, run in `lean/` with Lean 4.34.1 through a small timing wrapper (`leanchk`, which appends the last line below; the wrapper itself is not distributed) and recorded in `lean/lean_output.txt`, is

```
'A12.main' depends on axioms: [propext, Classical.choice, Quot.sound]
'A12.main_statement' depends on axioms: [propext, Classical.choice, Quot.sound]
'A12.zhang_zheng_fails_n13' depends on axioms: [propext, Classical.choice, Quot.sound]
leanchk: wall=124.31s maxrss=2369432kB exit=0
```

i.e. only the three standard axioms; in particular no `Lean.ofReduceBool`. The option `Elab.async false` is set so that the kernel checks run sequentially within the memory limit. An independent re-run of `lean Main.lean` with a freshly installed Lean 4.34.1 core toolchain (review of October 8, 2026) reproduced the three axiom lines exactly, with exit code 0, wall time 124.1 s and maximum resident set size 2.36 GB; the same review also reproduced the two search runs of Section 3 (seeds 77 and 41) bit for bit.

5 Limitations and open questions

- Novelty rests on a literature search (IACR ePrint, arXiv including 2025–2026 papers on odd-n Boolean functions, journal pages and survey slides); every source we could check that lists the known counterexamples [17, 16, 2] gives $n \in \{9, 11, 15, 21\}$ for odd n, and to our knowledge, as of October 8, 2026, the case $n = 13$ had not been settled. The conjecture is quoted from the original paper [19] in Section 1; the floor in the exponent of Theorem 1.1 follows [17].
- Only $n = 13$ is settled. For odd n the cases $n = 17, 19$ appear to remain open; our method extends in principle (the class $\mathrm{AGL}(1, p)$ for prime p) but kernel verification of $n = 17$ would be about 2^8 times more expensive.
- The minimum of Δ_f over balanced 13-variable functions is unknown; we found 112 but not 104 in $\mathrm{AGL}(1, 13)$. The nonlinearity 3980 of our function is below the bent concatenation bound 4032; whether $\Delta_f < 128$ is compatible with nonlinearity > 4032 for $n = 13$ is open.
- Degree 11, nonlinearity 3980, the $|r_f|$ distribution and the $\mathrm{AGL}(1, 13)$-invariance are script-checked only.

Remark 5.1 (Negative results on balanced nonlinearity). Before turning to the absolute indicator we looked for balanced functions beating the best known balanced nonlinearities 992 ($n = 11$) and 4036 ($n = 13$) [11, 13, 3], starting from $h(y) \oplus f(x)$ with f one of the published 9-variable functions of nonlinearity 242 [12]. An exhaustive search found that toggling 4 bits of any of 69 such 11-variable bases never gives a balanced function of nonlinearity ≥ 994, and tabu search over 8-bit toggles of the 13-variable bases (the route to 4036 in [11]) never approached 4038. These computations are not certified and are recorded only as negative evidence.

A Truth table

$f(x)$ is bit x (weight 2^x) of the following 8192-bit integer, written in hexadecimal with the most significant digit first, 64 digits per line (the same digits as `data/truth_table.hex` and the literal `tt` in `lean/Main.lean`):

```
ffeafdcdeba3a0e7e8cf885b8914b96efd80b4afd1d436dac5c60734de9228e9
eff6d450ce35c8bee247f7205e69a3d8a137e438517b5b64f3b8861c1cd4a882
e8aafe78e3253354a1ad0f67f4d1dafdb81d206beb7a0d0036e87987995ff3c1
8d171b6ef9255ac436122fce638b6861bb4ade94907d47a516e1f3758cc5815c
f9d18dc9faed2f84b91f1c671b1b6675dc1398f244bf3c2aeb75f743b298feb2
8bd112a30c543d9bf99e6ed844e240510b2cfcc47ed3802b828662affa0be013
c1f2463a538f78ecaed7586236c9a1705e6857480ceee0e8394a859b2c817d47
de8e65d8b2a8c220c3102af3642b98334669f842fa4e7b37c4a0e176840767b5
ef83b213c5b6f5c2fed9bce21cab85708e9317eb12a4693a539b478a2d3d6b73
```

a6b5170fc2d4be0c20749eea1fe01988ac9b7a62ee7a244fde0882c0eab9cb5d
c0dee64646188c5a55b536604ae7c28bbfc6d6b87ce9e2847570b94970112712
15cf1da4bfe4a0206bede31e8511498f945d9039381c88bfbbd804dfa905035b
a016af1c75295edc330fc4ae6ac0bcb1c8edf66f67d12c095a69a09298532b11
36e969c5666a648545e0b8eda814fdd15b83249d8027928e1da1c0077bb2756f
e3ac85b83c37e695da0c9880a14d4815b15f4301088def0b6821498bc2d45e5f
607c79d7fe857119bb8935b96f8e5e6fe071d914a8132f3d8131116f3d3fdb76
f8fb910b8b0d165af063de6cba36e409beeca7868bf0e91c17a4dddad4266e00
80ec935e133bedcf47099c3068d35fd9375ad3ca643a918958e64fb369df2b1e
dd39cb32576b55afe54ca7609ff944e41d412e75d6aca88d12eaed010283c185
9ca4d3ca2fc8691ce8f82a88583521fbe2bc4080940da145a9c8cad7a09e23b7
b145b6f8ed68613c25794294d5b0379d27738e620e68785025c8ac2be008859f
8eaaa42cf7688b817ae4e982fc59c5347b727a009b9635936f4546075c2f131d
0636f1ee53a288658bebbd21c9044d15699ba8a3b84b06fdd12703427183d4fb
923467a7c2401b875f9103b181918eef8edfe3d55175b2eb8dd60537155b77de
cd51136cc9ea46a16a321c9672fce2a00a5e55bbb1648dfd7cd8a451cab59f16
f594f9f3ba387daa792fe20348b1019362d838968c00c71d8680631b188f0617
5b6cb8963d86f4773d2828cc382180372022f811ca90edb6dcc05675bba3e357
26dad00b197192b79501492fd318d1ed46b38853f515152b2f9fdf493e377dfe
bd5a98a18466ce800ef55a3fec2d8776f6dd44f0d38185158c4620b720954327
da4672ef700a410700c584b6e8ef519f2cc15d477193d4dba50df77137f967fe
7c412aa02fc7a33eefacd1372a070793cacbc0930b268bd729fbd5ed33fd7dbe
e9003b16b2d703358985121f1dfb5ff690564f1713477dfe4eb75bfea7de3e68

B Orbit description

Equivalently, $f(x) = 1$ if and only if the $\mathrm{AGL}(1, 13)$-orbit of x has least element (as an integer) in

> 3, 11, 19, 23, 31, 39, 43, 55, 107, 127, 151, 183, 187, 191, 215, 223, 255, 303, 311, 367, 375, 383, 431, 443, 447, 471, 763, 767, 831, 895, 951, 1023, 1791, 2047, 4095, 8191.

These 36 orbits have sizes (in the same order) 78, 156, 52, 156, 78, 156, 156, 156, 39, 78, 156, 156, 156, 156, 156, 156, 78, 156, 52, 156, 156, 156, 156, 26, 156, 78, 156, 156, 156, 156, 39, 78, 52, 78, 13, 1, summing to 4096.

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